

A note on a question about polynomial multiplicities over $\mathbb{F}_2$

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Abstract

Bishnoi, Boyadzhyska, Das and Mészáros ask whether a polynomial over $\mathbb{F}_2$ that vanishes with multiplicity at least k at every nonzero point of $\mathbb{F}_2^n$, and with multiplicity at most $k - 1$ at the origin, has degree at least $2k - \lfloor k/2^{n-1} \rfloor$, when $k \geq 2^{n-2}$. We observe that the requested inequality is an immediate consequence of the multiplicity Schwartz–Zippel lemma of Dvir, Kopparty, Saraf and Sudan. The deduction works for all positive integers n, k , without the stated restriction on k .

1 The question and the standard estimate

All degrees below are ordinary total degrees of formal polynomials. For a nonzero polynomial $P \in \mathbb{F}[x_1, \dots, x_n]$ and $a \in \mathbb{F}^n$, write

$$P(a + z) = \sum_{\alpha \in \mathbb{Z}_{\geq 0}^n} c_\alpha z^\alpha.$$

Its multiplicity at a is

$$\text{mult}_a(P) = \min\{|\alpha| : c_\alpha \neq 0\}, \quad |\alpha| = \alpha_1 + \dots + \alpha_n.$$

This is Hasse multiplicity: all Hasse derivatives of total order less than $\text{mult}_a(P)$ vanish at a . We set $\text{mult}_a(0) = \infty$. In particular, a nonvanishing value has multiplicity zero. We do not reduce polynomials modulo the field equations, since that operation need not preserve multiplicities.

In the published version of *Subspace coverings with multiplicities*, Bishnoi, Boyadzhyska, Das and Mészáros [1, Question 4.3] ask the following: for positive integers n, k with $k \geq 2^{n-2}$, if $P \in \mathbb{F}_2[x_1, \dots, x_n]$ satisfies

$$\text{mult}_a(P) \geq k \quad (a \in \mathbb{F}_2^n \setminus \{\mathbf{0}\}), \quad \text{mult}_{\mathbf{0}}(P) \leq k - 1, \tag{1}$$

must one have

$$\deg P \geq 2k - \left\lfloor \frac{k}{2^{n-1}} \right\rfloor? \tag{2}$$

The question is numbered 5.2 in the earlier arXiv version [2]. The distinction matters because Question 4.3 in that version concerns a different covering problem.

The relevant estimate is already available in the work of Dvir, Kopparty, Saraf and Sudan [3, Definition 2.2 and Lemma 2.7].

Lemma 1 (Multiplicity Schwartz–Zippel). *Let $\mathbb{F}$ be any field, $n \geq 1$, and let $P \in \mathbb{F}[x_1, \dots, x_n]$ be a nonzero polynomial of total degree d . For every finite nonempty set $S \subseteq \mathbb{F}$,*

$$\sum_{a \in S^n} \text{mult}_a(P) \leq d |S|^{n-1}. \quad (3)$$

There is no restriction relating d to the field size in this lemma.

2 The deduction

Proposition 1. *For every pair of positive integers n, k , every polynomial $P \in \mathbb{F}_2[x_1, \dots, x_n]$ satisfying (1) satisfies (2).*

Proof. The condition at the origin excludes the zero polynomial. Put $d = \deg P$ and apply Lemma 1 with $S = \mathbb{F}_2$. Since multiplicities are nonnegative and $\mathbb{F}_2^n$ has $2^n - 1$ nonzero points,

$$k(2^n - 1) \leq \sum_{a \in \mathbb{F}_2^n} \text{mult}_a(P) \leq d 2^{n-1}.$$

Dividing by 2^{n-1} and using integrality of d gives

$$d \geq \left\lceil 2k - \frac{k}{2^{n-1}} \right\rceil = 2k - \left\lfloor \frac{k}{2^{n-1}} \right\rfloor.$$

□

Thus the polynomial inequality in the published Question 4.3 has an affirmative answer, without the assumption $k \geq 2^{n-2}$. The upper bound on the origin’s multiplicity is used here only to ensure that $P \neq 0$. In fact, the same argument applies to any nonzero polynomial with the required multiplicities at the other points.

The purpose of this note is to record this direct implication of the existing multiplicity Schwartz–Zippel lemma. It does not supply a new proof of that lemma or a new method for lower bounds, and it does not address the other covering questions in [1].

Assistance disclosure. The literature comparison, identification of the implication, and drafting of this note were carried out with substantial assistance from GPT-6 Astra in Codex, under the author’s direction. The source comparison and deduction are recorded in the [Noemesis research repository](#).

References

- [1] A. Bishnoi, S. Boyadzhiyska, S. Das and T. Mészáros. Subspace coverings with multiplicities. *Combinatorics, Probability and Computing* **32**(5) (2023), 782–795. [doi:10.1017/S0963548323000123](https://doi.org/10.1017/S0963548323000123).
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- [3] Z. Dvir, S. Kopparty, S. Saraf and M. Sudan. Extensions to the method of multiplicities, with applications to Kakeya sets and mergers. *SIAM Journal on Computing* **42**(6) (2013), 2305–2328. [doi:10.1137/100783704](https://doi.org/10.1137/100783704).